

# Performance Analysis Report: Offensive Language Classification

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## 1. Executive Summary

This research performs a thorough assessment of machine learning systems built to evaluate multilingual toxic comment classification. This analysis investigates both conventional methods comprising Logistic Regression and Random Forest alongside modern deep learning structures like Long Short-Term Memory (LSTM) networks and transformer-based models (BERT).

The standard classification performance assessment included specific strategies to tackle the class imbalance problems found in the data. The study provides performance understanding and strategic directions for enhanced model developments.

### Reported Accuracy

- **Logistic Regression:** 79.5%
- **Random Forest:** 74%
- **LSTM:** 80%
- **BERT:** Accuracy not reported; evaluated using precision and recall.

## 2. Model Performance Evaluation

### 2.1 Accuracy & Metrics

| Model               | Precision | Recall | F1-Score | AUC-ROC |
|---------------------|-----------|--------|----------|---------|
| Logistic Regression | 0.3967    | 0.5448 | 0.4591   | 0.6938  |
| Random Forest       | 0.2375    | 0.2836 | 0.2585   | 0.5554  |
| LSTM                | 0.78      | 0.80   | 0.75     | 0.59    |
| Bert                | N/A       | N/A    | 0.30     | 0.60    |

### Key Insights

- **Logistic Regression** achieved moderate performance, with recall outperforming precision.
- **Random Forest** yielded the weakest results across all metrics, highlighting limitations in handling linguistic context.
- **LSTM** offered the strongest overall performance, demonstrating superior balance between precision, recall, and F1-score.
- **BERT** exhibited high recall, indicating strong detection of toxic content, but precision and stability require further optimization.

## 3. Data Challenges

### Class Imbalance

- The dataset exhibited significant skew: **90.6% non-toxic** vs. **9.4% toxic**.

- **Undersampling** was employed to rebalance the distribution for binary classification.
- Additional labels (e.g., *abusive*, *menace*, *bigotry*) were also imbalanced; however, they were retained to support multi-label learning, preserving the dataset's richness and complexity.

## 4. Recommendations

To enhance model performance—particularly for underrepresented classes and multilingual content—the following strategies are recommended:

- **Threshold Optimization & Focal Loss:** Adjust decision thresholds and employ loss functions designed to penalize misclassification of minority classes to improve recall without excessively degrading precision.
- **Adoption of Multilingual Transformers:** Leverage pre-trained models such as **XLM-R** or **mBERT**, which are specifically optimized for multilingual inputs, thus eliminating the need for translation-based preprocessing.
- **Data Augmentation Techniques:** Utilize methods such as **SMOTE** or back-translation to synthetically enhance minority class samples, thereby improving learning stability and generalization.
- **Model Ensembling:** Explore ensemble approaches combining LSTM and BERT outputs to capitalize on complementary strengths and improve robustness.

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## 5. Conclusion

Among the models evaluated, LSTM and BERT showed the greatest potential for effective toxic language detection in a multilingual context. Although traditional models like Logistic Regression provide useful baselines, they lack the contextual sensitivity needed for nuanced classification tasks. Future work should focus on addressing data imbalance, utilizing multilingual transformer models, and implementing advanced training strategies to enhance classification accuracy and fairness across all classes.